**Project Design Phase**
**Solution Architecture**

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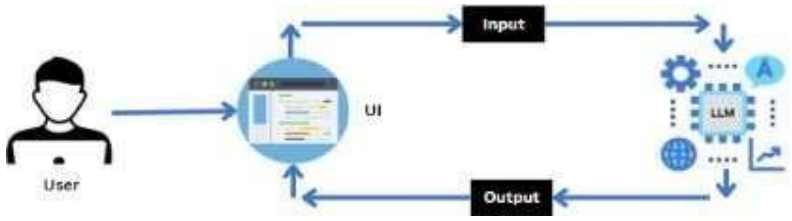
| Date | 18 February 2026 |
| Team ID | LTVIP2026TMIDS52544 |
| Project Name | Explore with AI: Custom Itineraries for Your Next Journey |
| Maximum Marks | 2 Marks |

## Solution Architecture:




## 1. Overview

Explore with AI is a web and mobile-based platform that allows users to enter travel preferences (destination, duration, interests) and receive a **personalized, AI-generated travel itinerary**. The system is designed for simplicity, speed, and scalability.

## 2. High-Level Components

### A. User Interface (Frontend)

- **Technology:** Streamlit (Python) or React.js for web/mobile interface
- **Functionality:**
  - Input fields for destination, trip duration, and preferences/interests
  - Submit button to generate itinerary
  - Display of AI-generated itinerary in a structured, day-wise format
  - Options to review, customize, or export itinerary (PDF/Text)
  - Display user-friendly error messages

### B. API Layer / Backend

- **Technology:** Python (Flask/FastAPI)
- **Functionality:**
  - Receives user input from frontend

    - Validates inputs (destination, dates, preferences)
    - Handles requests to the AI model
    - Returns structured itinerary results to frontend

### C. AI Engine

- **Technology:** Gemini Pro LLM / OpenAI GPT / other generative AI model
- **Functionality:**
    - Processes input parameters (destination, duration, interests)
    - Generates detailed day-wise itinerary including:
        - Attractions and sightseeing
        - Local dining recommendations
        - Suggested activities and timing
        - Travel tips and optional hidden gems
    - Ensures structured output for easy rendering on frontend

### D. Data Layer (Optional)

- **Technology:** Cloud database (MongoDB/PostgreSQL)
- **Functionality:**
    - Stores predefined destination data (optional)
    - Stores user-generated itineraries (if history/export feature is implemented)
    - Logs analytics for system monitoring

### E. Export & Reporting Module

- **Functionality:**
    - Convert itinerary to PDF or downloadable formats
    - Copy/export to clipboard
    - Optional integration with email or messaging for sharing

## 3. Key Features of Architecture

- **Modular Design:** Frontend, backend, AI engine, and data layer are loosely coupled.
- **Scalable:** Can handle more destinations, multiple users, and AI requests simultaneously.
- **Extensible:** Future features can include multi-destination planning, travel booking integrations, or content generation for blogs.
- **Simple Workflow:** Focused on **input → AI processing → output**, minimizing user friction.